



AKTU

B.Tech II-Year



Cyber security



Unit-2 in One Shot

Pdf Notes + PYQs + Imp. Questions

AKTU QUESTIONS

1.	Explain wireless devices with example. What are security challenges faced by wireless devices	AKTU 2023-24
2.	Explain the security measures and policy taken for mobile devices	AKTU 2023-24
3.	State some attack in mobile devices .what are security implication for organization	AKTU 2023-24
4.	Explain the different types of security policy that an organization should implement	AKTU 2022-23
5.	How does MFA work on mobile devices	AKTU 2022-23
6.	How does user detect and respond to phishing attack on their mobile devices	AKTU 2023-24
7.	What are the key component of a comprehensive mobile security policies for organsition	AKTU 2023-24

Wireless Devices

- Wireless devices are electronic gadgets that connect to other devices or networks without using wires or cables.
- They use signals like Wi-Fi, Bluetooth, or mobile data (4G/5G) to communicate.

Examples of Wireless Devices:

- Smartphones ,Tablets, Laptops with Wi-Fi,
- Wireless printers , Smart-watches,
- Wireless earphones
- Wi-Fi routers

Security Challenges Faced by Wireless Devices:

Challenge	Explanation
1. Unauthorized Access	Hackers can break into a device if it's not password-protected or properly secured.
2. Data Interception	Data sent over Wi-Fi or Bluetooth can be captured by attackers
3. Malware Attacks	Wireless devices can get infected through fake apps or unsafe websites.
4. Weak Encryption	If data is not encrypted (locked), hackers can easily read it during transmission.
5. Rogue Wi-Fi Networks	Attackers set up fake Wi-Fi hotspots to steal data when users connect to them.

Importance of mobile and wireless devices

1.Easy Communication:

You can call, message, or video chat with anyone anytime, anywhere.

2.Access to Internet:

These devices let you use the internet without wires – for browsing, social media, online shopping, and more.

3.Work from Anywhere:

People can work remotely using mobile devices – attending meetings, sending emails, or completing tasks from home or while traveling.

4.Entertainment:

You can watch movies, listen to music, play games, and read books – all on one device.

5.Online Services:

Banking, studying, shopping, and health checkups are now possible from your mobile without visiting any place.

6.Emergency Use:

They are very helpful in emergencies to call help or get quick information like maps or weather alerts.

Proliferation of Mobile and Wireless Devices

- The proliferation of mobile and wireless devices refers to the rapid increase in the use of smartphones, tablets, laptops, and other wireless gadgets across the world.
- These devices have become an essential part of daily life due to their ability to offer instant communication, internet access, and various services on the go.

Mobile Computing is “taking a computer and all necessary files and software out into the field.” Many types of mobile computers have been introduced since 1990s. They are as follows:

1. Portable Computer

- A regular computer that can be carried from one place to another.
- Needs AC power and setup before use.

Proliferation of Mobile and Wireless Devices

2. Tablet PC

- Looks like a notebook
- No keyboard – uses touchscreen and stylus.
- Good for drawing or reading.
- Can do most tasks a laptop can do.

3. Internet Tablet

- Looks like a tablet, but has less computing power.
- Cannot replace a normal computer.
- Used for web browsing, chatting, watching videos, and viewing pictures.
- Has limited apps

4. Personal Digital Assistant (PDA)

- A small, pocket-sized device.
- Works with a desktop computer to manage contacts, notes, emails, etc.
- Has limited features.

5. Ultra-Mobile PC

- A small, portable computer like a PDA.
- Runs a full operating system (like Windows).
- Can do many things a normal PC can do but in a small size.

6. Smartphone

- A PDA + mobile phone combined.
- Can make calls, send messages, browse the web, use apps, and much more.
- Very powerful and widely used today.

SOME TERMS

Term	Simple Meaning
Mobile Computing	Using a device while moving , like a phone or laptop in a car or train
Wireless Computing	Using the internet without wires , like through Wi-Fi or mobile data
Hand-held Devices	Small devices that fit in your hand like smartphones, PDAs, or tablets

How Users Can Detect and Respond to Phishing Attacks on Mobile Devices

How to Detect Phishing Attacks:

1. Suspicious Links or Emails:

- If you get a message or email from an unknown sender asking to click a link — be careful.
- Example: “Your bank account is locked, click here to unlock.”

2. Strange App Behavior:

- If an app asks for unusual permissions (like access to your contacts or SMS), it may be malicious.

3.Urgent Messages:

Messages with scary warnings (like “Your account will be closed today”) often trick users into clicking links.

4.Spelling and Grammar Mistakes:

Fake messages often have bad grammar or spelling errors.

5.Fake Websites:

Phishing sites look like real ones but have slightly different URLs (e.g., paypa1.com instead of paypal.com)

How Users Can Detect and Respond to Phishing Attacks on Mobile Devices

How to respond Phishing Attacks:

1. Do Not Click Links or Download Attachments

Avoid tapping on any suspicious links or attachments in SMS, emails, or messaging apps.

2. Do Not Enter Personal Information

Never enter passwords, banking details, or OTPs on suspicious websites or apps.

3. Report the Message

Report the phishing SMS or email to your mobile provider or email service (e.g., Gmail has “Report phishing” option).

Forward phishing SMS to **1909** in India (TRAI spam reporting).

4. Delete the Message

After reporting, delete the message immediately to avoid accidental clicking later.

How MFA(Multi-Factor Authentication)system work on mobile devices

Login Process:

Step 1: Enter your username and password
(Something you know)

Step 2:

- Receive a code by SMS,
- OR get a code from an authenticator app,
- OR confirm identity using fingerprint/face ID (Something you have or are)

You need to pass both steps to access your account.

Method	How it Works
SMS Code	A 6-digit code is sent to your mobile number
Authenticator App	App like Google Authenticator generates a code
Biometric Login	Uses fingerprint or face ID
Push Notification	App sends a “Yes/No” approval request to your phone

- Protects accounts even if someone knows your password
- Stops hackers from logging in without your device

Explain security measures taken for mobile devices

Measure	Explanation
1. Screen Lock	Use PIN, password, fingerprint, or face ID to lock the device.
2. Antivirus Software	Install trusted security apps to block viruses and malware.
3. Encryption	Data is stored in a coded form so others can't read it.
4. Remote Wipe	Allows users to erase data if the phone is lost or stolen.
5. Two-Factor Authentication (2FA)	Adds extra protection for apps and accounts.
6. Regular Updates	Keep your device and apps updated to fix bugs and security holes.
7. App Permissions	Give apps only the permissions they need (like access to camera, contacts).
8. Avoid Public Wi-Fi	Public networks are unsafe. Use VPN or mobile data for sensitive work.

Policy	Explanation
1. Bring Your Own Device (BYOD)	Rules set by companies when employees use personal phones for work.
2. Acceptable Use Policy	Guidelines on what users can or cannot do on their devices.
3. Data Backup Policy	Ensures important data is regularly saved to avoid loss.
4. App Installation Policy	Allows only approved and safe apps to be installed on the device.
5. Lost/Stolen Device Policy	Steps to report and secure data if a device is lost or stolen.
6. Access Control Policy	Only authorized users can access certain data or systems from mobile.

Key Components of Comprehensive Mobile Security for an Organization

Component	Explanation
1. Device Security	Use of screen locks (PIN, fingerprint, face ID) and full device encryption.
2. Mobile Device Management (MDM)	Software to monitor, manage, and secure mobile devices used in the organization.
3. Secure App Management	Only approved apps are allowed; unsafe apps are blocked or removed.
4. Network Security	Use of VPNs, secure Wi-Fi, and blocking untrusted networks.
5. Data Protection	Ensures data encryption, backups, and remote wipe if device is lost.
6. Access Control	Only authorized users can access company apps or data using login and 2FA.

Key Components of Comprehensive Mobile Security for an Organization

Component	Explanation
7. User Awareness & Training	Educating employees to avoid phishing, weak passwords, and unsafe downloads.
8. Policy Enforcement	Enforcing company rules like BYOD, acceptable use, and lost device reporting.
9. Regular Software Updates	Keeping OS and apps up to date to fix security flaws.
10. Threat Detection & Response	Tools to detect malware or hacking attempts and take quick action.

Attack Type	Explanation
1. Phishing	Fake messages or emails trick users into giving personal info or passwords.
2. Malware	Harmful apps or files that steal data, spy on users, or damage the device.
3. Spyware	Secretly monitors user activity (like calls, messages, or location).
4. Man-in-the-Middle	Hacker intercepts communication between device and server (e.g., on public Wi-Fi).
5. App-based Attacks	Fake or infected apps that steal data or control device functions.
6. SIM Card Swapping	Hacker tricks mobile provider to get your number on their SIM, stealing your access.
7. Bluetooth Hacking	Hacker connects to your device through Bluetooth without permission.

Security Implications for Mobile Devices:

Implication	Meaning
1. Data Theft	Personal, financial, or company data can be stolen.
2. Loss of Privacy	Hackers can access private messages, photos, and location.
3. Financial Loss	Bank frauds, online payment thefts, or SIM-based OTP theft.
4. Unauthorized Access	Hackers can control device, make calls, or send messages.
5. Damage to Reputation	Company data leaks can harm trust and brand image.
6. Legal Issues	Data breaches may lead to fines or lawsuits under privacy laws.
7. Business Disruption	Attacks can stop work, delete data, or shut down systems.

Trend	Explanation
1. Mobile Cloud Computing	Apps and data are stored on the cloud, so they can be accessed from anywhere.
2. 5G Technology	Very fast mobile internet for faster downloads and better mobile performance.
3. Mobile Payment Systems	Services like Google Pay, PhonePe, UPI are replacing cash and cards.
4. Wearable Devices	Smartwatches and fitness bands connected to mobile apps.
5. Mobile Apps for Everything	From learning to shopping – everything is now app-based.
6. Remote Work / Work from Anywhere	Employees can work using mobile devices from home or while traveling.
7. Mobile Device Management (MDM)	Companies use MDM tools to control and secure mobile use.

Types of Mobility:

Mobility Type	Explanation
1. Personal Mobility	Movement of individuals using personal mobile devices (phones, tablets).
2. Enterprise Mobility	Use of mobile technology by businesses to support employees and processes.
3. Social Mobility	Access to services and information that help improve lifestyle and learning.
4. Geographic Mobility	Ability to use mobile services while moving across locations or countries.

Implications of Mobility:

Implication	Explanation
1. Security Risks	More devices mean more chances of hacking or data leaks.
2. Need for Strong Policies	Companies must manage device use and data safety carefully.
3. Dependence on Internet	Without a good internet connection, mobility services may not work.

What is Credit Card Fraud in Mobile and Wireless Computing?

Credit card fraud in mobile and wireless computing means stealing and misusing someone's credit card information through mobile phones, wireless networks, or online platforms

Method/techniques	Explanation
1. Phishing via SMS/Email	Fake messages ask users to enter credit card details on fake websites.
2. Malicious Apps	Some apps secretly steal credit card numbers when entered.
3. Public Wi-Fi Attacks	Hackers can capture payment data entered over unsecured Wi-Fi.
4. Fake Payment Gateways	Fraudsters create fake mobile sites that look like real payment sites.
5. SIM Swapping	Hacker gets access to your number and OTPs for online payments.
6. Spyware or Key-loggers	Malware records what you type, including card numbers and passwords.

What Do You Understand by Authentic Service Security?

- **Authentic Service Security** means ensuring that a service or system is real, trustworthy, and secure, and that only the right users can access it.

- **Authentic** = Genuine or Real

The service must not be fake or tampered with.

- **Service Security** = Protection of Online Services

Make sure no unauthorized person can misuse the service.

Main Goals:

- Verify user identity (e.g., login with password or fingerprint)
- Prevent hackers from accessing or changing the service
- Protect data shared between user and service (e.g., banking, emails)

Cryptographic Security for Mobile Devices

Cryptographic security is essential for protecting sensitive data and communications on mobile devices. It uses mathematical algorithms to secure data and ensure only authorized parties can access it.

Technique	Description	Example Use
Encryption	Converts data into unreadable form to prevent unauthorized access	AES used to encrypt files and messages
Digital Signatures	Ensures data integrity and authenticity	Verifying app developers or software updates
Hashing	Converts data into fixed-length values; used for integrity	Password storage, data integrity checks
Public Key Infrastructure (PKI)	Uses private and public keys to secure data	secure email
Key Exchange Protocols	Safely exchange cryptographic keys	Diffie-Hellman used in secure messaging apps

Devices.

RSA (Rivest-Shamir-Adleman) is a **public-key cryptographic algorithm** used to secure data. It is widely used in mobile devices for **secure communication, digital signatures, and authentication**.

How RSA Works (Basic Steps)

Component	Description
Public Key	Shared with others to encrypt data
Private Key	Kept secret by the device to decrypt data
Encryption	Sender uses the recipient's public key to encrypt a message
Decryption	Recipient uses their private key to decrypt the message

Only the person with the private key can decrypt what was encrypted with the public key.

Application	Description
Secure Messaging	RSA is used in apps like WhatsApp for key exchange before symmetric encryption
Mobile Banking & Payments	RSA helps verify users and secure financial transactions
App Authentication	Verifies that an app or update comes from a trusted developer
VPNs & HTTPS	RSA is used during handshakes to set up secure web and VPN connections

Challenges of RSA on Mobile Devices

Challenge	Reason
Performance	RSA is slower than symmetric encryption (like AES)
Key Size	Needs long keys (2048 bits or more) for strong security
Battery Usage	RSA operations are CPU-intensive and may drain battery

1.MALWARE ATTACK

Type	Description
Spyware	Secretly collects personal data like location, messages
Trojan	Disguised as legitimate app but performs harmful tasks
Ransomware	Locks the phone and demands payment to unlock

2.NETWORK ATTACK

Type	Description
Man-in-the-Middle (MITM)	Attacker intercepts communication between user and service
Wi-Fi Sniffing	Hacker monitors unencrypted data over Wi-Fi
Fake Wi-Fi Hotspots	A rogue Wi-Fi network mimicking a real one

3.SOCIAL ENGINEERING ATTACKS

Type	Description	Example
Phishing (via SMS/email)	Fake messages or links trick users to enter credentials	"Your bank account is blocked. Click here"
Vishing (voice phishing)	Calls pretending to be from bank or government	"Share your OTP to avoid penalty"
Smishing (SMS phishing)	Fraudulent SMS with malicious link	"You've won a prize! Click this link!"

4.APP-BASED ATTACKS

Type	Description	Example
Malicious Apps	Installed from third-party stores or links	Stealing contacts, camera access
Privilege Escalation	App gains more access than allowed	Accesses files without permission
App Cloning	Fake version of popular app	WhatsApp clone stealing messages

Security Implications of Mobile Devices for Organizations

1. Data Leakage

Risk: Sensitive business data can be leaked via messaging apps, email, or cloud sync.

Example: Employee sends confidential report through personal WhatsApp.

Implication: Loss of intellectual property or customer trust.

2. Unsecured Network Access

Risk: Employees access company systems over public Wi-Fi.

Example: Logging into company email on open airport Wi-Fi.

Implication: Increases risk of man-in-the-middle and data interception attacks.

3. Malicious Mobile Apps

Risk: Employees may install apps that contain malware or spyware.

Example: A flashlight app stealing contact details.

Implication: Corporate data accessed or stolen via infected device.

4. Weak Authentication

Risk: Devices with no PIN, weak passwords, or no biometric security.

Example: Lost phone gives access to company files.

Implication: Unauthorized data access and potential data breach.

5. Insider Threats

Risk: Employees intentionally misuse mobile devices.

Example: Copying sensitive documents and sharing with competitors.

Implication: Legal and financial damage.

6. Lost or Stolen Devices

Risk: Devices contain confidential emails, files, or credentials.

Example: Employee loses tablet with client database.

Implication: Exposure of sensitive customer or internal information.

7. Inadequate Mobile Device Management (MDM)

Risk:

No central system to control or monitor devices.

Example: IT cannot wipe data from a lost device.

Implication: Lack of control over corporate data on mobile endpoints.

How Organizations Can Reduce These Risks

Solution	Description
Mobile Device Management (MDM)	Monitor, secure, and control devices used for work
Encryption	Encrypt all mobile data at rest and in transit
Strong Authentication	Use biometrics, MFA, and strong passwords
App Control	Restrict app downloads to verified sources
Regular Updates	Force updates for OS and apps
Remote Wipe	Remotely delete data from lost/stolen devices
Employee Training	Educate staff about mobile threats and best practices

Security Policies

➤ A **security policy** is a formal set of rules and guidelines that govern how mobile devices should be used, managed, and protected within an organization.

Why is a Mobile Security Policy Important?

1. Protects Sensitive Business Data

Mobile devices often access **emails, databases, client info, and confidential files**.

A policy ensures this data is **encrypted, protected, and not leaked**.

2. Controls Device Access

The policy defines **who can use mobile devices**, what types are allowed (e.g., company-owned vs BYOD), and how they access company networks.

Prevents **unauthorized users** from gaining access.

3. Reduces Malware & App Threats

The policy can **restrict downloads** to official app stores.

Ensures use of **approved antivirus or security tools** on devices.

4 Secures Network

Communication

Enforces use of **VPNs** and **encrypted connections** when using public or external networks.

Prevents **man-in-the-middle** attacks.

5.Minimizes Legal and Compliance

Risks

Helps meet data protection laws (e.g., GDPR, HIPAA).

Prevents legal penalties from **data leaks** or **misuse** of personal information.

6.Provides Monitoring and Incident Response

Defines how to **monitor devices** and handle incidents like lost phones, breaches, or misuse.

Enables **remote wiping** of data on lost/stolen devices

Operating Guidelines for

Implementing Mobile Device Security Policies

1. Conduct Risk Assessment

- **Identify threats:** e.g., data leakage, lost/stolen devices, malicious apps.
- **Assess vulnerabilities** in mobile usage, device types, and user behavior.
- **Evaluate impact** on business in case of a breach

2. Define Scope of the Policy

Clarify:

Who the policy applies to (employees, contractors).

What devices are covered (BYOD, company-owned, tablets, smartphones).

What data and services can be accessed via mobile.

3. Implement Access Controls

- Enforce:
 - **Strong passwords/PINs**
 - **Biometric authentication**
 - **Multi-factor authentication (MFA)** for accessing company apps
- Limit access to only what users need (**principle of least privilege**).

4. Mandate Device Encryption

Require **full-disk encryption** on all mobile devices.

Ensure **data in transit** (emails, file uploads, etc.) is encrypted using **SSL/VPN**

5. App Management and Control

Whitelist approved apps and block risky or unverified apps.

Prohibit use of apps that **access sensitive data** unnecessarily.

Regularly update and patch apps to fix vulnerabilities.

Registry Settings for Windows Mobile Powered Devices

- The registry in Windows Mobile devices is a central database that stores configuration settings for the operating system, hardware, and installed applications.
- It works similarly to the Windows registry on desktop computers.
- The registry contains keys and values that control how the device behaves.
- It is used to customize system settings, manage startup programs, configure network settings, and control hardware features.
- Developers and advanced users can modify the registry to enable hidden features, tweak performance, or fix issues.
- Editing the registry is done using tools like PHM Registry Editor or Total Commander (with registry plugin).
- Incorrect changes to the registry can lead to system errors or crashes, so backups are recommended before editing.

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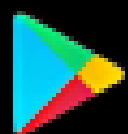


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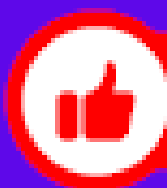


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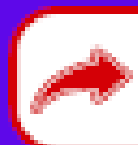
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